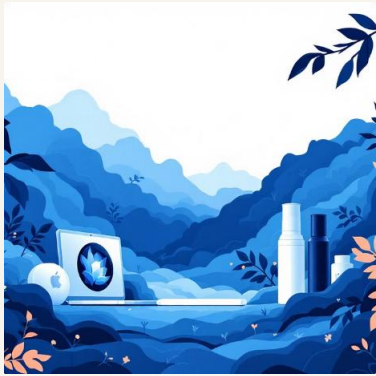




Machine Learning Portfolio Optimization

A comprehensive framework for portfolio optimization using tree-ensemble forecasting and advanced feature engineering across a 10-stock portfolio – **The Lose-Lose Fund Quant Approach**

Portfolio Composition & Data Foundation



Tech Giants

AAPL, MSFT, NVDA, GOOG,
CSCO



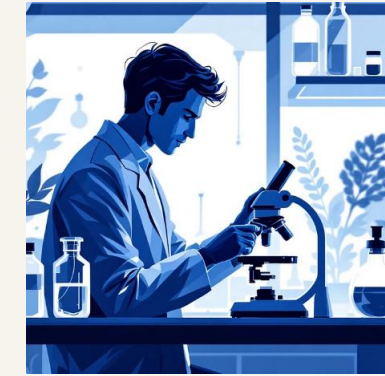
Energy Sector

CVX, XOM



Consumer Staples

KO, PG



Healthcare

PFE

10 years of historical closing price data analyzed with CBOE Volatility Index (VIX) as market-wide risk sentiment proxy

Comprehensive Feature Engineering

38 quantitative features engineered to capture market dynamics, momentum, and volatility patterns



Technical Momentum

* **10 x RSI-14** indicators capturing overbought/oversold conditions for mean reversion patterns

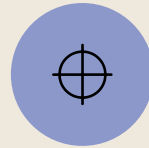
* Rolling sum momentum indicators

(3-day, 5-day, 10-day windows) representing cumulative return persistence over rolling time horizons

$$Prtf_Momentum_t^{window} = \sum_{i=t-w+1}^t r_i$$

Market Risk Proxy

VIX Z-score signal with $\pm 2\sigma$ bands providing binary fear/greed indicators



Trend Signals

10 EMA crossover signals (3-day vs 10-day) identifying bullish/bearish momentum shifts

Autoregressive Patterns Momentum Proxies

Portfolio Lags Indicators of 1-day, 2-day and 3-day Lags to catch the Autoregressive Patterns.



Volatility Measures

10 volatility spike ratios detecting regime changes in realized volatility

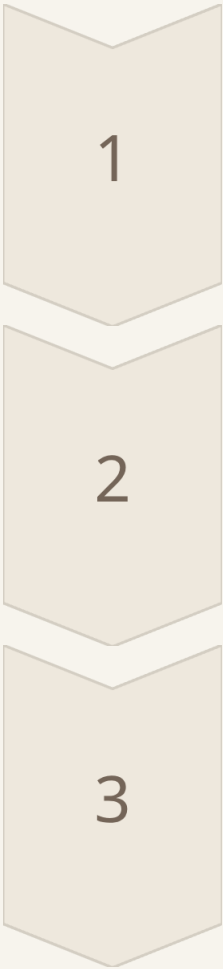
$$\sigma_{t=2} / \sigma_{t=10}$$

Cross-Sectional Dynamics

Herding Index measuring coordinated movement across portfolio assets

$$Herd_t = \frac{1}{N} \sum_{i=1}^N s_{i,t}$$

Model Selection: Three Ensemble Methods



Random Forest

Bootstrap aggregation creating decorrelated tree predictions through random subsets

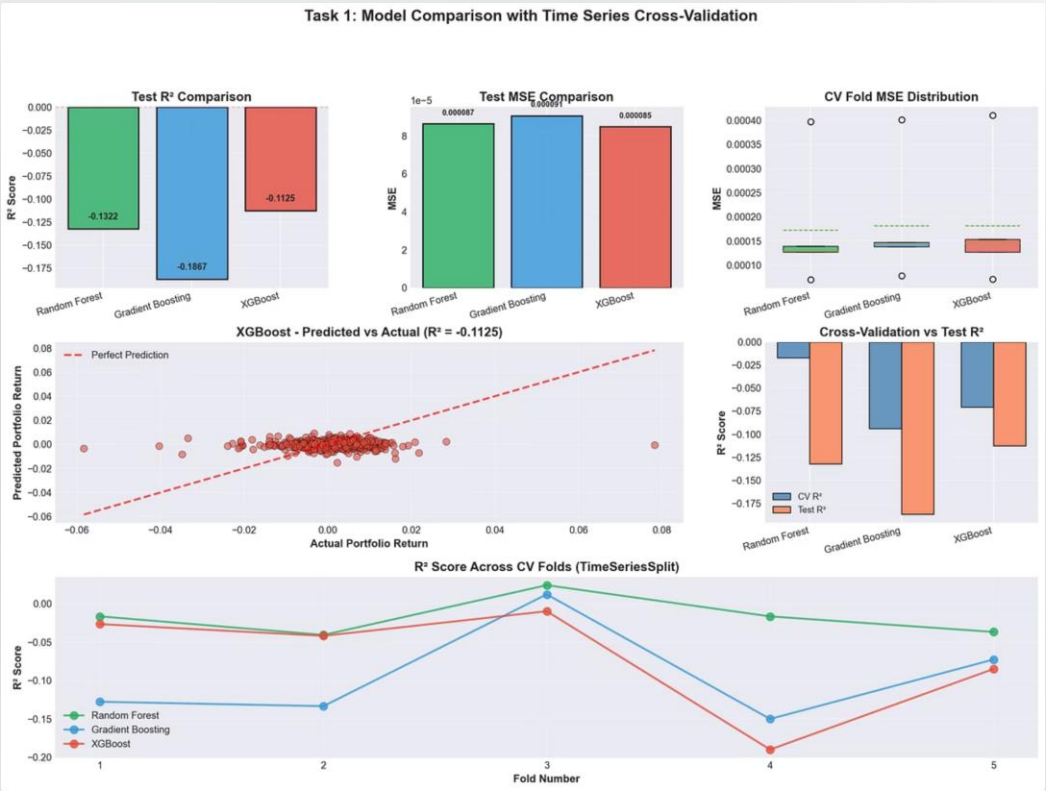
Gradient Boosting

Sequential ensemble fitting residual errors with iterative bias reduction

XGBoost

Enhanced regularization with L1/L2 penalties and sophisticated pruning algorithms

Rigorous evaluation using 5-fold time series cross-validation with MSE, R^2 , and MAE metrics



XGBoost Emerges as Winner

Statistical comparison via paired t-tests on fold-level MSE revealed XGBoost delivered the best hold-out performance despite all models showing limited linear explanatory power at daily frequency.

Test MSE

0.00008501

Lowest error among all models

Test R^2

-0.1125

Best out-of-sample fit

```
=====
STATISTICAL COMPARISON (Paired t-tests on CV Folds)
=====
Random Forest vs Gradient Boosting:
  • t-statistic: -3.6833
  • p-value: 0.021142
  • Winner: Random Forest **
  • Result: Significantly better (p < 0.05)

Random Forest vs XGBoost:
  • t-statistic: -2.1395
  • p-value: 0.099145
  • Winner: Random Forest *
  • Result: No significant difference (p ≥ 0.05)

Gradient Boosting vs XGBoost:
  • t-statistic: 0.0473
  • p-value: 0.964557
  • Winner: XGBoost
  • Result: No significant difference (p ≥ 0.05)

=====
TASK 1 SUMMARY TABLE
=====
```

	Model	Train R^2	Test R^2	Test MSE	Test MAE	CV MSE
	Random Forest	0.343817	-0.141432	0.00008693	0.00586729	0.00017253
	Gradient Boosting	0.837698	-0.131409	0.00008617	0.00611117	0.00018046
	XGBoost	0.696073	-0.088966	0.00008294	0.00611769	0.00018026

```
=====
📄 Saved: task1_model_comparison.csv
🏆 Best Model: XGBoost (Test  $R^2$  = -0.088966)
=====
```

XGBoost is an optimized gradient boosting implementation with additional regularization. Key hyperparameters to tune:

- n_estimators & learning_rate:**
 - Similar to Gradient Boosting
 - Lower learning_rate → More trees needed
 - We test: [50, 100, 200] × [0.01, 0.05, 0.1]
- max_depth:**
 - Controls tree complexity
 - XGBoost typically uses shallower trees (3-6) than RF
 - We test: [3, 5, 7, 10]
- min_child_weight:**
 - Similar to min_samples_leaf
 - Higher → More conservative, prevents overfitting
 - We test: [1, 3, 5]
- subsample (Row Sampling):**
 - Fraction of samples used per tree
 - <1.0 → Introduces randomness, prevents overfitting
 - We test: [0.7, 0.8, 0.9]
- colsample_bytree (Column Sampling):**
 - Fraction of features used per tree
 - <1.0 → Decorrelates trees, improves generalization
 - We test: [0.7, 0.8, 0.9]

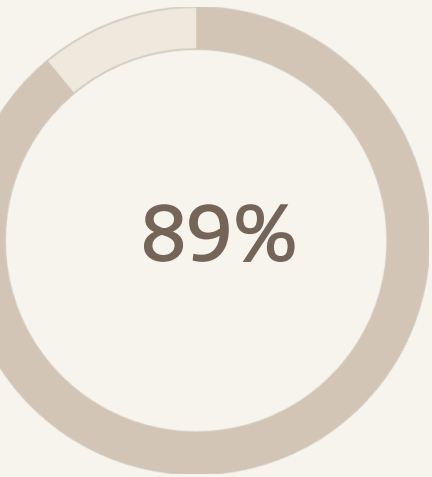
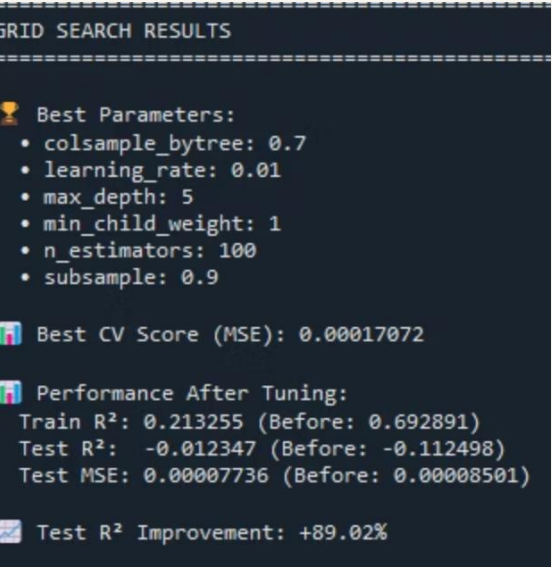
Variance-dominated learning task typical at daily frequency sets stage for careful regularization via hyperparameter tuning

Hyperparameter Optimization

01

Grid Search Configuration

972 unique parameter combinations tested across 5 critical hyperparameters



Improvement in Test R² after tuning

02

Exhaustive Evaluation

4,860 individual model fits (972 combinations × 5 folds) ensuring thorough exploration

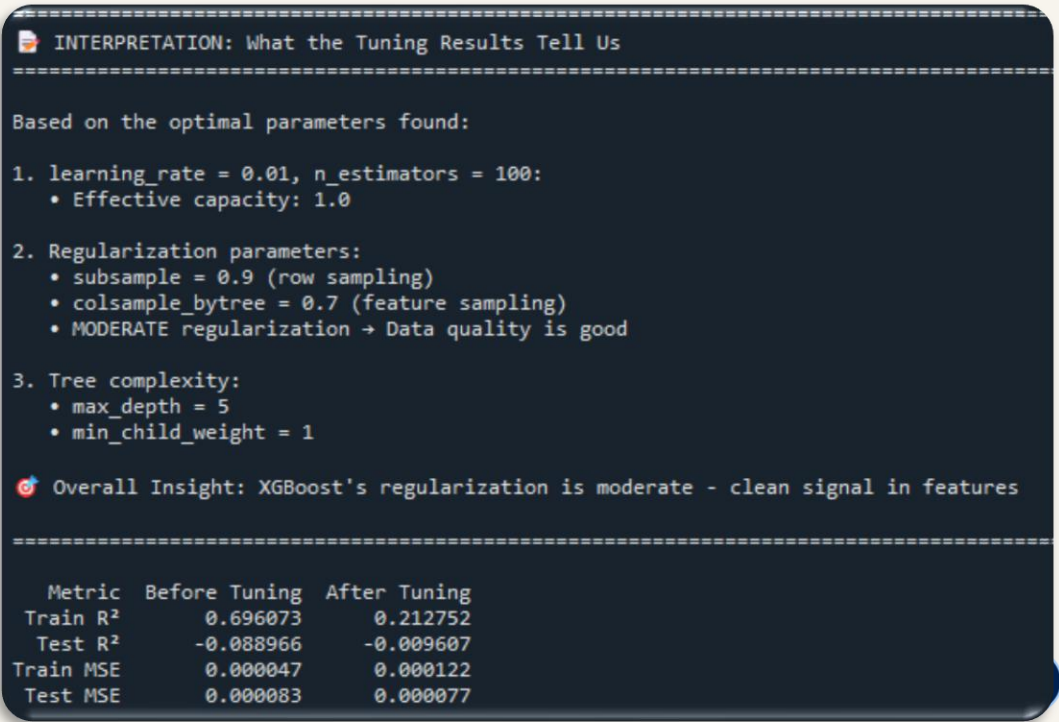


Training R² reduced from 0.69, mitigating overfitting

03

Optimal Configuration

Learning rate 0.01, 100 estimators, max depth 5, subsample 0.9, colsample 0.7



Feature Importance Analysis

Built-In Importance

Microsoft features showed highest predictive power at 12.6% total importance, measuring contribution to reducing impurity during tree splits

✓ Top 15 Features (Built-in Importance):

Feature	Importance
Portfolio_Lag2	0.055908
Vol_Spike_AAPL	0.055770
Vol_Spike_MSFT	0.053253
EMA_Cross_MSFT	0.051115
Herding_Index	0.048663
Portfolio_Lag3	0.045770
Portfolio_Momentum_3d	0.045121
RSI_14_XOM	0.044290
Vol_Spike_CVX	0.043564
Vol_Spike_PFE	0.041665
Portfolio_Lag1	0.040013
Portfolio_Momentum_10d	0.038846
RSI_14_PFE	0.035352
Portfolio_Momentum_5d	0.033233
RSI_14_NVDA	0.030189

FEATURE CATEGORY BREAKDOWN

Category-wise Importance Summary:

• RSI

: 0.2536 (25.4%)

• EMA_Cross

: 0.1334 (13.3%)

• Vol_Spike

: 0.3054 (30.5%)

• VIX

: 0.0000 (0.0%)

• Herding

: 0.0487 (4.9%)

• Portfolio_Lag

: 0.1417 (14.2%)

• Portfolio_Momentum

: 0.1172 (11.7%)

🏆 Most Important Stock: MSFT (Total Importance: 0.1259)

This stock's features collectively have the highest predictive power.

SHAP Analysis

Game-theoretic marginal contributions revealed top drivers: **Herding Index, lagged portfolio returns, Vol Spikes in MSFT & PG**

✓ Top 10 Features Comparison:

Feature	Built-in (norm)	SHAP (norm)
Herding_Index	0.870399	0.999999
Vol_Spike_PG	0.373973	0.962199
Vol_Spike_MSFT	0.952497	0.762254
Portfolio_Lag2	1.000000	0.593886
Portfolio_Lag1	0.715689	0.583238
RSI_14_KO	0.327014	0.582015
Portfolio_Lag3	0.818669	0.476208
RSI_14_CSCO	0.285712	0.412317
Portfolio_Momentum_3d	0.807051	0.393389
Portfolio_Momentum_5d	0.594415	0.351027

📊 Correlation between Built-in and SHAP: 0.5011

🏆 Top 5 Features (SHAP-ranked):

Feature	SHAP	Built-in
Herding_Index	0.000114	0.048663
Vol_Spike_PG	0.000110	0.020908
Vol_Spike_MSFT	0.000087	0.053253
Portfolio_Lag2	0.000068	0.055908
Portfolio_Lag1	0.000067	0.040013

Moderate but significant correlation ($p \sim 0.50$) between methods validates feature selection and model interpretation

Key Predictive Insights

Momentum Effects – Portfolio Lags Play Role

Lagged returns validate autocorrelation over 1-3 day horizons due to market microstructure and information diffusion

Volatility Regimes

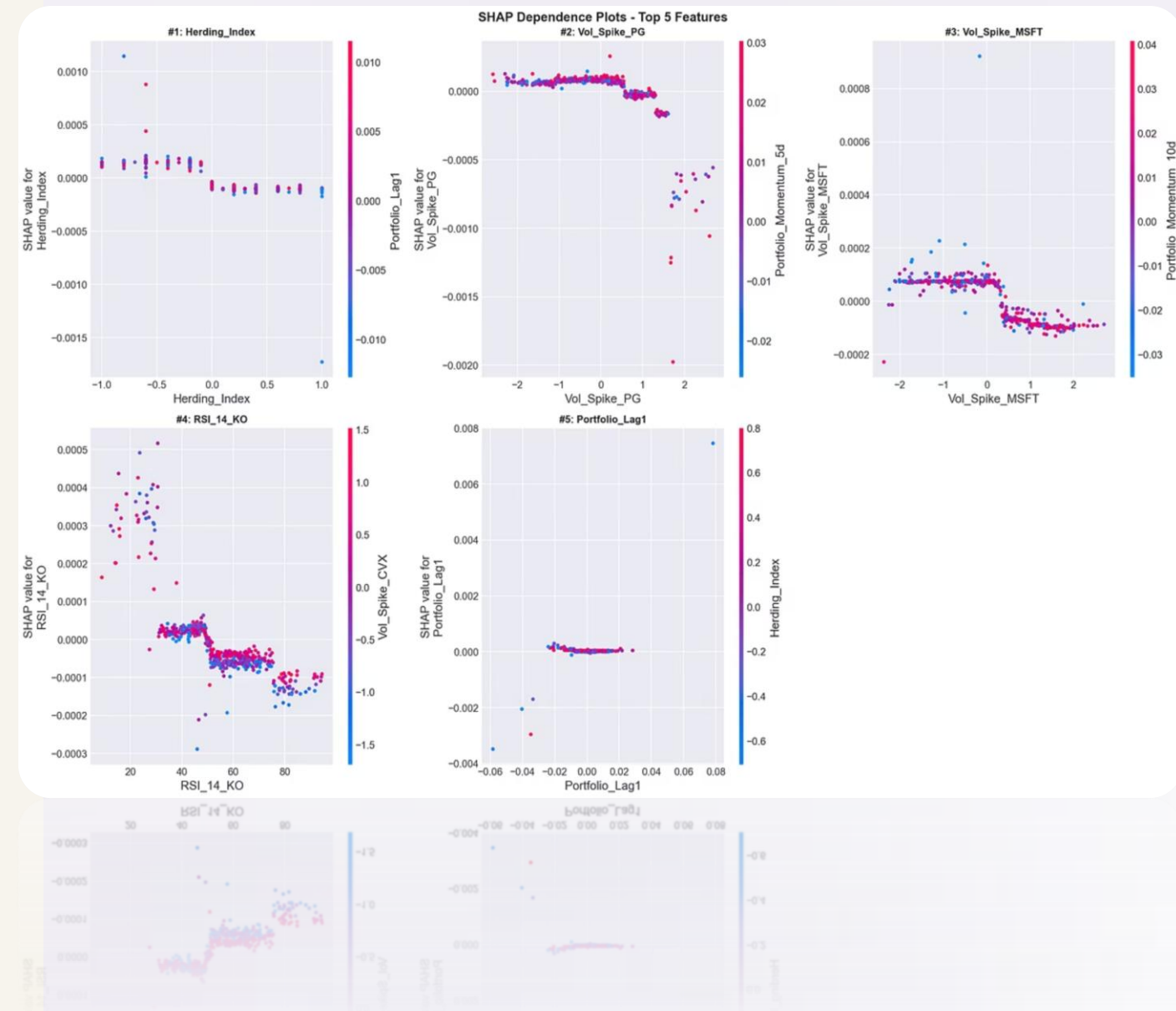
Spike ratios detect when individual stock volatility diverges from norms, predicting portfolio-wide movements

High-Tech Dominance and Non-Cyclical Power

MSFT, NVDA, AAPL from Tech giants along with KO and PG exhibit stronger feature importance, reflecting higher beta, volatility, and market-moving potential along with predictive power of the Non-Cyclical Stocks

Herding Dynamics

Herding dynamics further influence expected portfolio returns by reflecting the degree of Systemic coordination across assets.



Portfolio Strategy Comparison

Rolling-window backtesting with 30-day windows evaluated three strategies over 500 trading days test period

Equal-Weighted

21.82% annual return, 13.9% volatility, 1.57 Sharpe ratio

Model-Based

19.4% annual return, 14.8% volatility, lower Sharpe performance

Optimized Sharpe

25.7% annual return, 14.92% volatility, **1.72 Sharpe ratio**

CALCULATING PERFORMANCE METRICS

PERFORMANCE COMPARISON TABLE

	Equal-Weighted	Model-Based	Optimized Sharpe
AnnRet(%)	21.822639	19.432288	25.706337
AnnVol(%)	13.877060	14.781903	14.917976
Sharpe	1.572569	1.314600	1.723179
MaxDD(%)	-16.933500	-18.356582	-15.973175
WinRate(%)	57.000000	56.800000	55.200000
TotalRet(%)	51.245381	43.865610	62.859087

🏆 Best Strategy: Optimized Sharpe (Sharpe = 1.72)

🕒 Best Sharpe Ratio: Optimized Sharpe (Sharpe = 1.72)

AnnRet(%)	21.822639	19.432288	25.706337
AnnVol(%)	13.877060	14.781903	14.917976

Outstanding Results

1.72

Sharpe Ratio

Best risk-adjusted
performance achieved

25.7%

Annual Return

Superior to
benchmark strategies

62.9%

Total Return

Lose-Lose Fund

Beat The market

89%

R² Improvement

Through hyperparameter
optimization

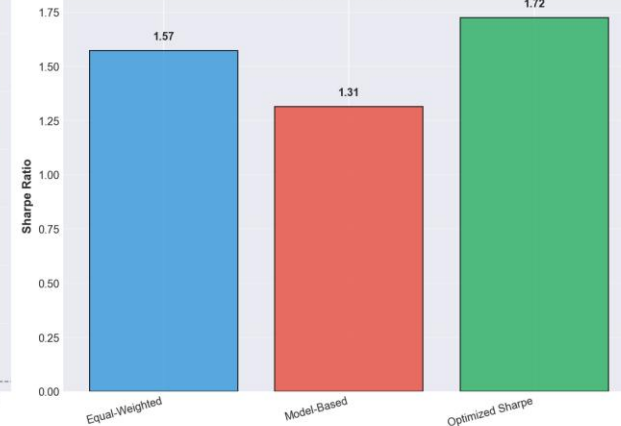
The Lose-Lose Fund's quant trading team beat the market by combining ML-driven expected returns with mean-variance optimization, demonstrating meaningful economic gains over static allocation approaches

Task 4: Portfolio Optimization & Backtesting Results

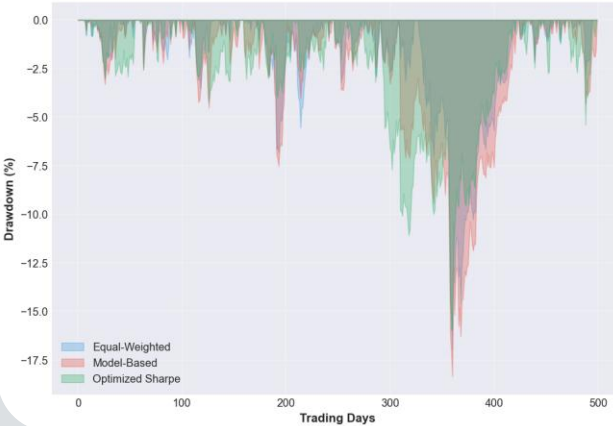
Cumulative Returns



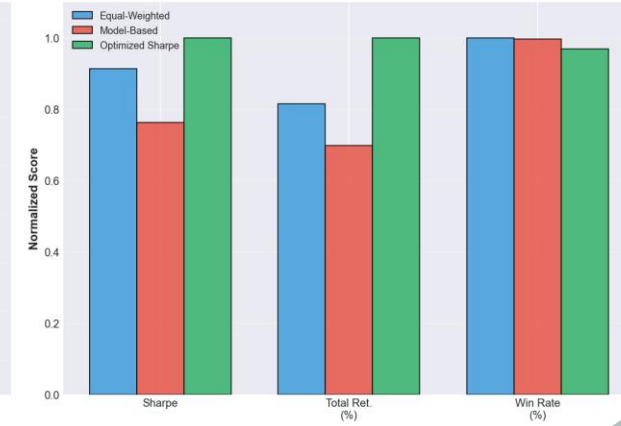
Sharpe Ratio Comparison



Drawdown Over Time



Performance Metrics Comparison (Normalized)



**RULE No. 1
NEVER LOSE MONEY.**

**RULE No. 2
NEVER FORGET
RULE NO. 1**

The Lose-Lose Investment
Fund tries to lose your money
–*but* with no success!

LOSE WITH US!

LOSE WITH US!

Thank U!

